

991/ 981, MY 2012-

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000000

Diagnostic information PDK control unit

Multiplexer sticks

Diagnostic conditions

- Drive in all gears

Possible fault causes

- ♦ Dirt particle temporarily stuck in the hydraulic moved switching valve (Multiplexer)
- ♦ switching valve (Multiplexer) sticks permanently
(part of hydraulic control unit in the gearbox)

Fault setting condition(s)

- Wrong moved shift rod

Fault effects

- ♦ a yellow CheckEngine Warning will appear at the instrument cluster and desired gears can't be engaged
⇒ causes a flushing of the Multiplexer through alternately hydraulically switching
- If the fault is still active after 5 flushings:
 - ♦ the depending gears will be blocked

P0562

Diagnosis information PDK control unit

Supply voltage

Diagnostic conditions

- Ignition on
- with overvoltage Engine running

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Generator faulty
- ♦ Battery faulty

Fault setting condition(s)

- Supply voltage < 7 V
- Supply voltage > 16.5 V
- Supply voltage > 7... 9 V

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ All pressure regulators and solenoid valves are switched off
- ♦ Hydraulic control is switched off
- ♦ Gear held
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ No emergency program diagnostics
- ♦ One-touch operation and Sport program are deactivated
- ♦ Start enable possible only in position P
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ Pressure-sensor plausibility check is switched off
- ♦ Pressure control is switched off
- ♦ Stop message from CAN is contradicted

- ◆ Substitute wheel speed

Control unit setpoints

- ◆ Supply voltage at terminal 30 $9.1\text{ V} \Rightarrow 16.4\text{ V}$
- ◆ Voltage at terminal 15 $9.5\text{ V} \Rightarrow 16.0\text{ V}$

P0563

Diagnosis information PDK control unit

Supply voltage

Diagnostic conditions

- Ignition on
- with overvoltage Engine running

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Generator faulty
- ♦ Battery faulty

Fault setting condition(s)

- Supply voltage < 7 V
- Supply voltage > 16.5 V
- Supply voltage > 7... 9 V

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ All pressure regulators and solenoid valves are switched off
- ♦ Hydraulic control is switched off
- ♦ Gear held
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ No emergency program diagnostics
- ♦ One-touch operation and Sport program are deactivated
- ♦ Start enable possible only in position P
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ Pressure-sensor plausibility check is switched off
- ♦ Pressure control is switched off
- ♦ Stop message from CAN is contradicted

- ◆ Substitute wheel speed

Control unit setpoints

- ◆ Supply voltage at terminal 30 $9.1\text{ V} \Rightarrow 16.4\text{ V}$
- ◆ Voltage at terminal 15 $9.5\text{ V} \Rightarrow 16.0\text{ V}$

P0603

Diagnosis information PDK control unit

Data record invalid

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Incorrect or faulty data version
- ♦ PDK control unit faulty

Fault setting condition(s)

- Internal diagnostic routine is faulty

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ Emergency stop function

P0604

Diagnosis information PDK control unit

PDK control unit - internal fault

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ PDK control unit faulty

Fault setting condition(s)

- Internal diagnostic routine returns fault

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ Emergency stop function

P0605

Diagnosis information PDK control unit

Data record invalid

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Incorrect or faulty data version
- ♦ PDK control unit faulty

Fault setting condition(s)

- Internal diagnostic routine is faulty

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ Emergency stop function

P0614

Diagnosis information PDK control unit

Data version does not match engine

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Wrong data version for DME (DFI) control unit
- ♦ Wrong data version for PDK control unit
- ♦ PDK control unit programmed incorrectly
- ♦ PDK control unit coded incorrectly - the data record does not match the vehicle

Fault setting condition(s)

- The flashed data version does not match the engine

Fault effects

- ♦ Transmission fault display in instrument cluster control unit
- ♦ Emergency stop function
- ♦ All hydraulic control elements are switched off
- ♦ All low-side electric pressure adjusters are switched off
- ♦ All low-side solenoid valves are switched off
- ♦ No R gear
- ♦ No Start/Stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ Ban one-touch operation and Sport program
- ♦ Start enable only in position P
- ♦ Electric pressure adjusters 5 and 6 are depressurized and multiplexer is changed over to locked shift rod
- ♦ Pressure control is switched off

Fault finding notes

- ♦ Program the correct data version for the engine

P0702

Diagnosis information PDK control unit

PDK control unit - internal fault

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ PDK control unit faulty

Fault setting condition(s)

- Internal diagnostic routine returns fault

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ Emergency stop function

P0706

Diagnosis information PDK control unit

Selector lever position information implausible

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Selector lever control unit faulty

Fault setting condition(s)

- Value for the position of the selector lever not valid

Fault effects

- ♦ None

P0711

Diagnosis information PDK control unit

Transmission temperature sensor implausible

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Wiring harness and plug faulty
- ♦ Temperature-dependent internal resistance (internal-resistance test)
- ♦ Transmission temperature sensor faulty

Fault setting condition(s)

- Transmission fluid temperature is not within an upper and lower limit of the electronic pressure adjuster temperature

Fault effects

- ♦ Electrical substitute value is assumed
- ♦ Substitute value is assumed
- ♦ All adaptations are deactivated
- ♦ No start/stop

Temperature-sensor setpoints

Temperature	Resistance [Ohm]
♦ 20 °C (68 °F)	903 ... 1,003
♦ 40 °C (104 °F)	1,057 ... 1,157
♦ 60 °C (140 °F)	1,226 ... 1,326
♦ 80 °C (176 °F)	1,380 ... 1,480

P0712

Diagnosis information PDK control unit

Transmission temperature sensor - short circuit to ground

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit in line between oil temperature sensor and PDK control unit
- ♦ Oil temperature sensor faulty

Fault setting condition(s)

- Measured voltage < threshold value (0.5 V)

Fault effects

- ♦ Substitute value is used
- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ All adaptations are deactivated
- ♦ No start/stop

P0731

Diagnosis information PDK control unit

Transmission ratio monitoring - gear train half 1

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Hydraulics faulty
- ♦ Transmission faulty

Fault setting condition(s)

- Transmission ratio monitoring - gear train half 1

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ Driving with gear train half 2 (only 2nd, 4th, 6th and R gears possible)
- ♦ Moving off in all gears
- ♦ No start/stop
- ♦ Torque limitation
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

Diagnosis information PDK control unit

Transmission ratio monitoring - gear train half 2

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Hydraulics faulty
- ♦ Transmission faulty

Fault setting condition(s)

- Transmission ratio monitoring - gear train half 2

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ Driving with gear train half 1 (only 1st, 3rd, 5th and 7th gears possible)
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P0754

Diagnostic information - PDK control unit

Solenoid valve 1 (selection of gear train half)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Solenoid valve 1 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjuster 4, 5, 6 is switched off
- ♦ Low-side solenoid valve 1, 2 is switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop mode
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function
- ♦ If short circuit to B+, then all hydraulic control elements are switched off
- Breakdown

P0759

Diagnostic information - PDK control unit

Solenoid valve 2 (shift cylinder selection)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Solenoid valve 2 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjuster 4, 5, 6 is switched off
- ♦ Low-side solenoid valve 1, 2 is switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop mode
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function
- ♦ If short circuit to B+, then all hydraulic control elements are switched off
- Breakdown

P0761

Diagnostic information PDK control unit

Multiplexer sticks

Diagnostic conditions

- Drive in all gears

Possible fault causes

- ♦ Dirt particle temporarily stuck in the hydraulic moved switching valve (Multiplexer)
- ♦ switching valve (Multiplexer) sticks permanently
(part of hydraulic control unit in the gearbox)

Fault setting condition(s)

- Wrong moved shift rod

Fault effects

- ♦ a yellow CheckEngine Warning will appear at the instrument cluster and desired gears can't be engaged
⇒ causes a flushing of the Multiplexer through alternately hydraulically switching
- If the fault is still active after 5 flushings:
 - ♦ the depending gears will be blocked

Diagnostic information PDK control unit

Multiplexer sticks

Diagnostic conditions

- Drive in all gears

Possible fault causes

- ♦ Dirt particle temporarily stuck in the hydraulic moved switching valve (Multiplexer)
- ♦ switching valve (Multiplexer) sticks permanently
(part of hydraulic control unit in the gearbox)

Fault setting condition(s)

- Wrong moved shift rod

Fault effects

- ♦ a yellow CheckEngine Warning will appear at the instrument cluster and desired gears can't be engaged
⇒ causes a flushing of the Multiplexer through alternately hydraulically switching
- If the fault is still active after 5 flushings:
 - ♦ the depending gears will be blocked

Diagnostic information PDK control unit

Multiplexer sticks

Diagnostic conditions

- Drive in all gears

Possible fault causes

- ♦ Dirt partical temporarily stuck in the hydraulic moved switching valve (Multiplexer)
- ♦ switching valve (Multiplexer) sticks permanently
(part of hydraulic control unit in the gearbox)

Fault setting condition(s)

- Wrong moved shift rod

Fault effects

- ♦ a yellow CheckEngine Warning will appear at the instrument cluster and desired gears can't be engaged
⇒ causes a flushing of the Multiplexer through alternately hydraulically switching
- If the fault is still active after 5 flushings:
 - ♦ the depending gears will be blocked

Diagnostic information PDK control unit

Multiplexer sticks

Diagnostic conditions

- Drive in all gears

Possible fault causes

- ♦ Dirt partical temporarily stuck in the hydraulic moved switching valve (Multiplexer)
- ♦ switching valve (Multiplexer) sticks permanently
(part of hydraulic control unit in the gearbox)

Fault setting condition(s)

- Wrong moved shift rod

Fault effects

- ♦ a yellow CheckEngine Warning will appear at the instrument cluster and desired gears can't be engaged
⇒ causes a flushing of the Multiplexer through alternately hydraulically switching
- If the fault is still active after 5 flushings:
 - ♦ the depending gears will be blocked

P0771

Diagnostic information PDK control unit

Multiplexer sticks

Diagnostic conditions

- Drive in all gears

Possible fault causes

- ♦ Dirt particle temporarily stuck in the hydraulic moved switching valve (Multiplexer)
- ♦ switching valve (Multiplexer) sticks permanently
(part of hydraulic control unit in the gearbox)

Fault setting condition(s)

- Wrong moved shift rod

Fault effects

- ♦ a yellow CheckEngine Warning will appear at the instrument cluster and desired gears can't be engaged
⇒ causes a flushing of the Multiplexer through alternately hydraulically switching
- If the fault is still active after 5 flushings:
 - ♦ the depending gears will be blocked

P0772

Diagnostic information PDK control unit

Multiplexer sticks

Diagnostic conditions

- Drive in all gears

Possible fault causes

- ♦ Dirt particle temporarily stuck in the hydraulic moved switching valve (Multiplexer)
- ♦ switching valve (Multiplexer) sticks permanently
(part of hydraulic control unit in the gearbox)

Fault setting condition(s)

- Wrong moved shift rod

Fault effects

- ♦ a yellow CheckEngine Warning will appear at the instrument cluster and desired gears can't be engaged
⇒ causes a flushing of the Multiplexer through alternately hydraulically switching
- If the fault is still active after 5 flushings:
 - ♦ the depending gears will be blocked

P0817

Diagnosis information PDK control unit

Control line for starter interlock - short circuit to ground, to B+ or open circuit

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ PDK control unit faulty
- ♦ DME control unit faulty

Fault setting condition(s)

- Measured voltage < threshold value for short circuit to ground (0.35 V)
- Threshold value for short circuit to ground (0.35 V) < measured low-side voltage < threshold for open circuit (1.8 V)
- Measured low-side voltage > threshold value for short circuit to B+ (3.0 V)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ No start/stop

P081A

Diagnosis information PDK control unit

Control line for starter interlock - short circuit to ground, to B+ or open circuit

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ PDK control unit faulty
- ♦ DME control unit faulty

Fault setting condition(s)

- Measured voltage < threshold value for short circuit to ground (0.35 V)
- Threshold value for short circuit to ground (0.35 V) < measured low-side voltage < threshold for open circuit (1.8 V)
- Measured low-side voltage > threshold value for short circuit to B+ (3.0 V)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ No start/stop

P081B

Diagnosis information PDK control unit

Control line for starter interlock - short circuit to ground, to B+ or open circuit

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ PDK control unit faulty
- ♦ DME control unit faulty

Fault setting condition(s)

- Measured voltage < threshold value for short circuit to ground (0.35 V)
- Threshold value for short circuit to ground (0.35 V) < measured low-side voltage < threshold for open circuit (1.8 V)
- Measured low-side voltage > threshold value for short circuit to B+ (3.0 V)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ No start/stop

Diagnosis information PDK control unit

Pressure sensor 1 - measured value implausible

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Hydraulics faulty
- ♦ Transmission faulty
- ♦ Pressure sensor 1 faulty

Fault setting condition(s)

- Pressure deviation of clutch 1 adapted > threshold (temperature-dependent characteristic)

-30 -10 0 10 45 90 °C (-22 14 32 50 113 194 °F)

1.75 1.55 1.45 1.35 1.25 1.25 bar

- Pressure deviation of clutch 1 not adapted > threshold (temperature-dependent characteristic)

-30 -10 0 10 45 90 °C (-22 14 32 50 113 194 °F)

2.25 2.05 1.75 1.5 1.5 1.5 bar

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 2 (only 2nd, 4th and 6th gears)
- ♦ No R gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ All adaptations are deactivated
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

Diagnosis information PDK control unit

**Pressure sensor 1, electrical fault
or
shadow fault**

INFORMATION

- ♦ From **software version Q033**, an electrical fault in pressure sensor 1 is indicated only via fault code **P0843**.
- ♦ From this software version, an operating state that could have been caused by brief sticking of one of the pressure adjusters is documented with fault entry **P181A**.
⇒ This leads to flushing of the pressure adjusters during driving, and this can cause one-time shift delays. However, there are no further impairments for the driver.

 There is generally no fault in this case.

♦ **Whether an electrical fault was present** can be read in the extended fault memory under > **Electrical fault if value = 1** <.

- If the value is **1**, an electrical fault was detected in pressure sensor 1.
 - ▶ Perform fault finding.
- There was no electrical fault if the value is **0**; only flushing was performed.
 - ▶ Erase fault memory.

Diagnostic conditions

- Ignition on
- Trip with different operating conditions

Possible fault causes

- **For P181A** ⇒ see information above
- **For P0843:**
 - ♦ Open circuit
 - ♦ Short circuit to B+
 - ♦ Short circuit to ground
 - ♦ Loose contact or corrosion on connectors or lines
 - ♦ Pressure sensor 1 faulty

Fault setting condition(s)

- Measured pressure > threshold value for short circuit to B+ (21 bar)
- Measured pressure < threshold value for short circuit to ground or open circuit (0.35 bar)

- If the faults relating to the oil temperature, pressure sensor 1 or pressure sensor 2 are entered, then the sensor ground is missing

Fault effects

- ◆ Transmission limp-home display in the instrument cluster control unit
- ◆ Driving with gear train half 2 (only 2nd, 4th and 6th gears)
- ◆ No R gear
- ◆ Moving off in all gears
- ◆ No start/stop
- ◆ Torque limitation
- ◆ No limp-home program diagnostics
- ◆ Deactivate all adaptations
- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

Diagnostic information - PDK control unit

Pressure sensor 2 - measured value implausible

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ If fault P18B8 is also entered as an active fault ⇒ Clutch faulty
- ♦ Hydraulics faulty
- ♦ Transmission faulty
- ♦ Pressure sensor 2 faulty

Fault setting condition(s)

- Pressure deviation of clutch 2 adapted > threshold (temperature-dependent characteristic)
-30 (-22) -10 (14) 0 (32) 10 (50) 45 (113) 90 (194) °C (°F)
1.75 (25) 1.55 (22) 1.45 (21) 1.35 (20) 1.25 (18) 1.25 (18) bar (psi)
- Pressure deviation of clutch 2 not adapted > threshold (temperature-dependent characteristic)
-30 (-22) -10 (14) 0 (32) 10 (50) 45 (113) 90 (194) °C (°F)
2.25 (33) 2.05 (30) 1.75 (25) 1.5 (22) 1.5 (22) 1.5 (22) bar (psi)

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Driving with gear train half 1 (only 1st, 3rd, 5th, 7th and Reverse gears possible)
- ♦ Moving off in all gears
- ♦ No Start/Stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ Deactivate all adaptations
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P0848

Diagnosis information PDK control unit

Pressure sensor 2 - short circuit to B+, to ground or open circuit

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure sensor 2 faulty

Fault setting condition(s)

- Measured pressure > threshold value for short circuit to B+ (21 bar)
- Measured pressure < threshold value for short circuit to ground or open circuit (0.35 bar)
- If the faults relating to the oil temperature, pressure sensor 1 or pressure sensor 2 are entered, then the sensor ground is missing

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 1 (only 1st, 3rd, 5th, 7th and R gears possible)
- ♦ Moving off in all gears
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ Deactivate all adaptations
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

Diagnosis information PDK control unit

Pressure sensor 3 (controlled transverse-differential lock) - measured value implausible

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Hydraulics faulty
- ♦ Transverse lock faulty
- ♦ Pressure sensor 3 faulty

Fault setting condition(s)

- Pressure deviation of clutch 1 adapted > threshold (temperature-dependent characteristic)

-30	-10	0	10	45	90 °C	(-22	14	32	50	113	194 °F)
1.75	1.55	1.45	1.35	1.25	1.25	bar					

- Pressure deviation of clutch 1 not adapted > threshold (temperature-dependent characteristic)

-30	-10	0	10	45	90 °C	(-22	14	32	50	113	194 °F)
2.25	2.05	1.75	1.5	1.5	1.5	bar					

Fault effects

- ♦ Controlled transverse-differential lock is deactivated
- ♦ Low-side electrical pressure adjuster 7 is switched off

Diagnosis information PDK control unit

Pressure sensor 3 (controlled transverse-differential lock) - short circuit to ground or open circuit

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure sensor 3 faulty

Fault setting condition(s)

- Measured pressure > threshold value for short circuit to B+ (21 bar)
- Measured pressure < threshold value for short circuit to ground or open circuit (0.35 bar)

Fault effects

- ♦ Controlled transverse-differential lock is deactivated
- ♦ Low-side electrical pressure adjuster 7 is switched off

Diagnosis information PDK control unit

Shiftlock solenoid - short circuit to ground, to B+ or open circuit

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Selector lever control unit faulty

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Shiftlock function can no longer be carried out
- ♦ The selector lever is locked in position P and can only be unlocked mechanically (emergency release)
- ♦ Hydraulic control 4 is switched off
- ♦ Low-side electrical pressure adjuster 7 is switched off
- ♦ Shiftlock solenoid is deactivated